LC - OSCILATORI

→ koristi se za generiranje sinusnih napona viših frekvencija do nekoliko stotina MHz

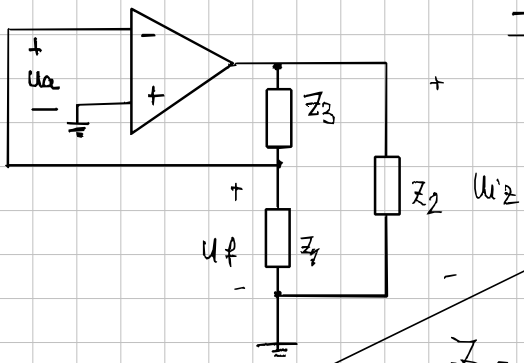
▸ FREKV. SELEKTIVNA POV. VEZA
· rezonantni krug sastavljen od kondenzatora i zavojnica

▸ zbog boljih VF svojstava → aktivne komponente za formiranje A-grane LC-oscilatora su diskretni bipolarni ili unipolarni tranzistori

Primjeri:
· Colpittsov oscilator
· Hartleyev oscilator
· Oscilator s kristalom

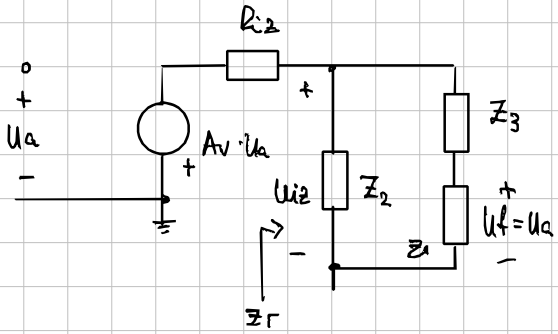
Opći oblik LC-oscilatora

spoj oscilatora



nadomjesna shema

A-grana β-grana

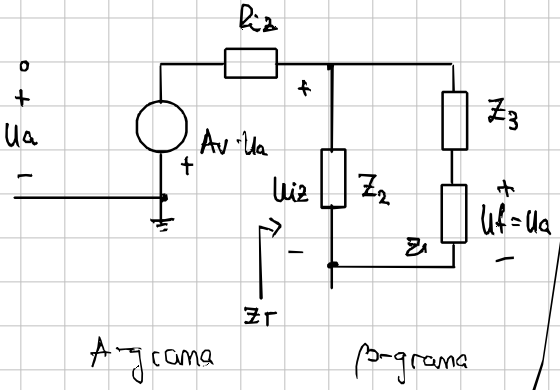


$$A_v = \frac{u_{iz}}{u_a} = -A_v \cdot \frac{Z_T}{Z_T + R_{iz}}$$

$$Z_T = Z_2 \parallel (Z_3 + Z_1) = \frac{Z_2(Z_3+Z_1)}{Z_2+Z_3+Z_1}$$

$$A_v = -A_v \cdot \frac{Z_2(Z_3+Z_1)}{Z_2+Z_3+Z_1} \cdot \frac{1}{\frac{Z_2(Z_3+Z_1)}{Z_2+Z_3+Z_1} + R_{iz}} = -A_v \cdot \frac{Z_2(Z_3+Z_1)}{Z_2(Z_3+Z_1) + R_{iz}(Z_2+Z_3+Z_1)}$$

$$\beta = \frac{u_f}{u_{iz}} = \frac{Z_1}{Z_1+Z_3} \Rightarrow \beta A_v = \frac{Z_1 \cdot Z_2}{R_{iz}(Z_1+Z_2+Z_3) + Z_2(Z_3+Z_1)}$$



$$\beta A_V = -A_V \frac{Z_1 \cdot Z_2}{R_{iz}(Z_1+Z_2+Z_3)+Z_2(Z_1+Z_3)}$$

$\Rightarrow Z_1 = jX_1 \qquad Z_2 = jX_2 \qquad Z_3 = jX_3$

$\hookrightarrow X = \frac{-1}{\omega C}$ ili $X = \omega L$

$$\beta A_V = +A_V \frac{X_1 \cdot X_2}{j R_{iz}(X_1+X_2+X_3) - X_2(X_3+X_1)}$$

Na frekvenciji oscilirajja $\omega_0 \longrightarrow \measuredangle \beta(\omega_0)\cdot A_V(\omega_0) = 0°$

$\Rightarrow$ pojačanje $\beta(\omega_0)\cdot A_V(\omega_0) > 0, \in \mathbb{R}$ ----> Imaginarni dio $= \emptyset$

$R_{iz} > 0 \longrightarrow X_1 + X_2 + X_3 = 0 \longrightarrow X_1 + X_3 = -X_2$

$$\Rightarrow \beta A_V = +A_V \frac{X_1 \cdot \cancel{X_2}}{-\cancel{X_2}\cdot(X_3+X_1)} = A_V \cdot \frac{X_1}{X_3+X_1}$$

$$\boxed{\beta A_V = -A_V \cdot \frac{X_1}{X_2}}$$

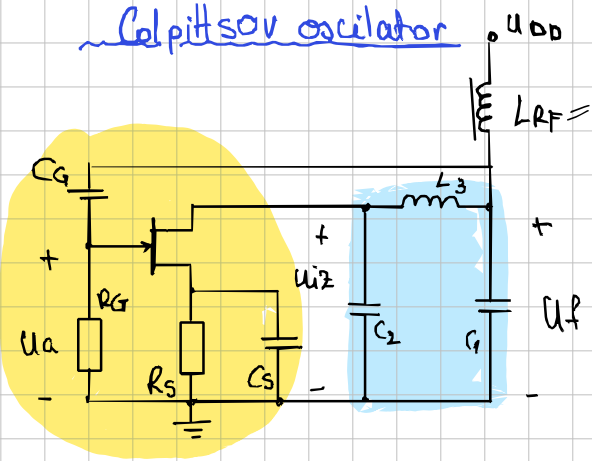
- X_1 i X_2 su istog predznaka, a X_3 je suprotnog predznaka
- Ako su Z_1 i Z_2 kapaciteti $\longrightarrow Z_3$ je induktivitet i obrnuto

$$|\beta(\omega_0)\cdot A_V(\omega_0)| = 1 \Longrightarrow A_{V\,min} = \frac{X_2}{X_1}$$

A-grana okreće fazu za 180° (minus stezaljka)

$\hookrightarrow$ β-grana treba još 180° okrenuti $\longrightarrow$ Uf mora ispasti $U_a \to U_f = U_a$

Colpittsov oscilator



$X_1 = \frac{-1}{\omega C_1}$ $\quad X_2 = \frac{-1}{\omega C_2}$ $\quad X_3 = \omega L_3$

$L_{RF} =$ → ne vodi na AE → dodam mag. materijal

Na frekvenciji oscilirajja ω_0

$$\frac{-1}{\omega_0 C_1} - \frac{1}{\omega_0 C_2} + \omega_0 L_3 = 0$$

$$-\frac{1}{C_1} - \frac{1}{C_2} + \omega_0^2 L_3 = 0$$

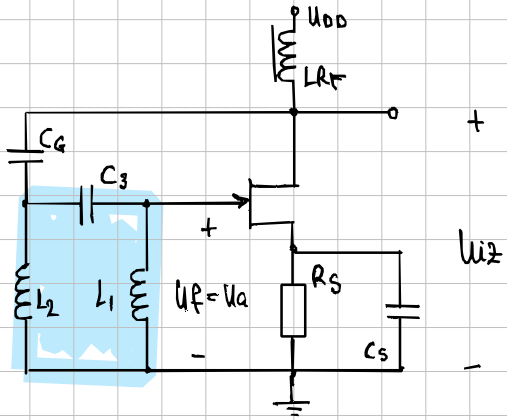
$$\omega_0 = \sqrt{\left(\frac{1}{C_1} + \frac{1}{C_2}\right)\frac{1}{L_3}}, \quad \omega_0 = 2\pi f$$

$$f_0 = \frac{1}{2\pi\sqrt{L_3}}\sqrt{\left(\frac{1}{C_1} + \frac{1}{C_2}\right)}$$

$$A_v = g_m \cdot r_d \longrightarrow g_m \geq \frac{C_1}{r_d \cdot C_2}$$

nemaju dod sklopove za stabilizaciju amplitude, ma ovamo izvana dodati

Hartleyev oscilator



$X_1 = \omega L_1$ $\quad X_2 = \omega L_2$ $\quad X_3 = \frac{-1}{\omega C_3}$

Na frekvenciji oscilirajja ω_0

$$\omega_0 L_1 + \omega_0 L_2 - \frac{1}{\omega_0 C_3} = 0$$

$$L_1 + L_2 - \frac{1}{\omega_0^2} \cdot \frac{1}{C_3} = 0$$

$$C_3 (L_1 + L_2) = \frac{1}{\omega_0^2} \longrightarrow \omega_0 = \frac{1}{\sqrt{(L_1+L_2)C_3}}$$

$$f_0 = \frac{1}{2\pi\sqrt{C_3(L_1+L_2)}}$$

$$A_v = g_m \cdot r_d \longrightarrow g_m \geq \frac{L_2}{r_d \cdot L_1}$$

Piezoelektrični kristal

→ Želimo povećati titrajnu kvalitetu, dobrotu (L_3)

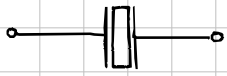
↳ faktor dobrote L_3 odredujemo kvalitetu postavljanja frekvencije

⇒ ŽELIMO MALI SERIJSKI OTPOR

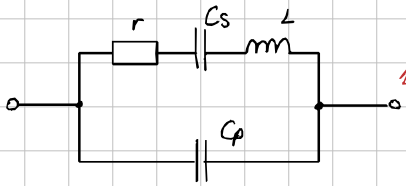
· Istosmjerni napon uzrokuje u piezoelektričnom kristalu naprezanja i deformacije

↳ mehaničke vibracije ⟶ kristal oscilira na svojoj mehaničkoj rezonantnoj frekvenciji ⟶ mehaničke vibracije proizvode na krajevima kristala izmjenični napon

⇒ kristal se električki ponaša kao rezonantni RLC-krug vrlo malog serijskog otpora, odnosno visokog faktora dobrote



električni simbol



(cijeli sklop)

nadomjesni sklop

· Zanemarujući mali otpor r iz nadomjesne shema:

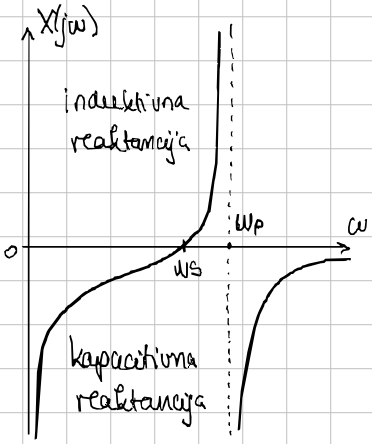
$$Z(j\omega) = X_p \parallel (X_s + X_L) = \frac{1}{j\omega C_p} \parallel \left(\frac{1}{j\omega C_s} + j\omega L\right)$$

$$Z(j\omega) = \frac{\frac{1}{j\omega C_p}\left(\frac{1}{j\omega C_s} + j\omega L\right)}{\frac{1}{j\omega C_p} + j\omega L + \frac{1}{j\omega C_s}}$$

$$Z(j\omega) = \frac{-j}{\omega C_p} \cdot \frac{\omega^2 - \omega_s^2}{\omega^2 - \omega_p^2}$$

$$\omega_s = \frac{1}{\sqrt{L C_s}}$$

$$\omega_p = \frac{1}{\sqrt{L \frac{C_s C_p}{C_s + C_p}}}$$



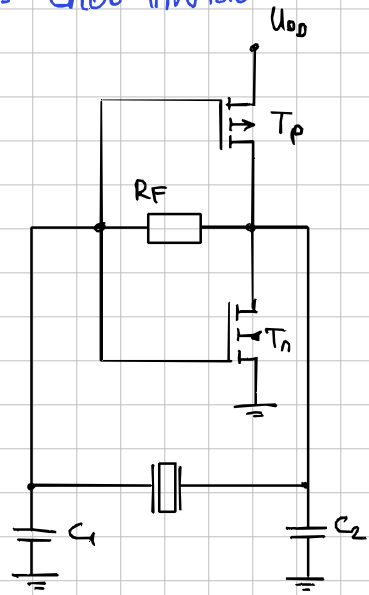
Rezonantne frekvencije ⟶ $\omega_p > \omega_s$

$C_s \ll C_p$ ⟶ ω_p i ω_s razlikuju se za manje od 1%

Oscilatori s kristalom

Zamjenom zavojnice L_3 u Colpittsovom oscilatoru s kristalom

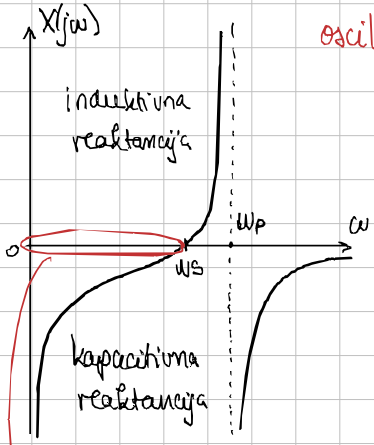
Piercevov oscilator s CMOS invertorom



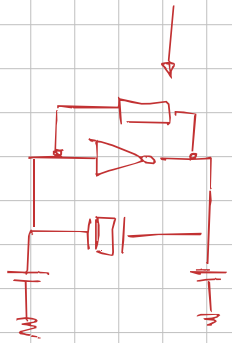
$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC_s} + \frac{1}{L}\cdot\frac{1}{C_p + \frac{C_1 C_2}{C_1 + C_2}}}$$

$$f_0 \approx \frac{1}{2\pi\sqrt{LC_s}} = \frac{\omega_s}{2\pi}$$

tj. ovo je ta posljedica zamjene L_3 sa cijelim ovim sklopom



oscilaciju imamo



Impedancija je idealno nula → max prijenos signala i najbolji prijenos energije